

Lesson 30 — Capstone Expo (Design Challenge)

Matches LEGO Teacher Portal lesson: "Password Pose".

Objective (I can...): I can design and present a complete machine that senses, decides, tracks data, reuses functions, and responds to an AI model I trained. **CS concept:** Integrating classifiers with events, conditionals, variables, and functions · **Builds on:** Lessons 1-29 — the whole course. *Privacy: on-device in the browser, no login, images not saved. Also serves as flex/catch-up — use today to finish and polish anything outstanding from the AI unit.*

AI Connection (2 min): Today's build senses, decides, counts, reuses your own functions, and responds to an AI YOU trained and evaluated — every unit of the year working as one system. *First-principles question:* "Which idea from this year is doing the most work in your capstone, and why?"

Materials: Device per pair (Teachable Machine + Coding Canvas), LEGO CS & AI kit, plan sheets, props, notebooks. **Model already built** — reuse the AI responder model (L26) or a small reconfig; the 30 min is plan + code + test + polish, not build-from-scratch.

60-minute flow - Warm-up (8): Capstone pitch — pairs state in one sentence what their machine does and which trained category drives it. *(Theme: Capstone expo.)* - **Concept + demo (10):** Challenge: build an interactive project where a trained classifier drives a LEGO build that also uses at least one conditional and at least one variable or function. Coach models connecting a recognition to a behavior and restates the privacy points. - **Build/Code (30):** Pairs plan and code the integrated capstone on the already-built model — classifier output → condition → a function that acts, with a variable tracking something meaningful. *Swap at 15 min.* - **Challenge (7):** Add a branch so a different recognition leads to a different outcome, or a count that changes the behavior over time. - **Share + pack-up (5): Capstone expo** — each pair demos their machine live while the coach records a short video clip for the parent note; audience names one concept they see working. **Photograph each build.** Close Teachable Machine tabs (nothing saved).

Pair roles: Co-plan; Driver codes / Navigator directs the demo and cues recognitions; swap at 15 min. *(Trio rotates plan/code/present.)*

Differentiation: *Support* — one recognition, one conditional, one function; partial plan. *Extension / fast-finisher* — multiple recognitions, branching outcomes, a variable, and a parameterized function. *Reteach* — reuse their Lesson 27-28 trained model and add one new behavior.

Exit check: Pair delivers a working machine that uses a trained, evaluated classifier to drive behavior, integrating a conditional and a variable or function, and can explain how it works. **Progress layer:** Send home the **Unit 6 (course finale) parent note** ("your student can train and evaluate an AI model and build a machine that senses, decides, tracks data, and reuses its own functions") with the **build photo** and the short demo clip — the year's capstone. **At-home (optional):** Write a half-page "engineer's summary" of your capstone: what it does, how the AI drives it, and one thing you'd improve with more time. ~10 min.

Standards Crosswalk

Aligned to the **CSTA K-12 CS Standards, grades 6-8** — primarily **Algorithms & Programming (2-AP)**, with real connections to **Data & Analysis (2-DA)** and **Impacts of Computing (2-IC)**. This is a targeted map, not a claim of full coverage. *Maps to the Virginia CS Standards of Learning, grades 6-8 (confirm exact SOL numbers).*

Unit (Lessons)	Skills taught	CSTA grades 6-8 codes targeted
1 • Basics (1-5)	Algorithms, sequencing, decomposition, flowcharts, pseudocode, abstraction, iterative design, building to spec	2-AP-13 (decompose problems into parts); 2-AP-10 (use flowcharts/pseudocode to design algorithms); 2-AP-17 (test and refine); 2-AP-15 (seek and use feedback — at the spec review)
2 • Loops (6-10)	Count, forever, and nested loops; loop variable; controlled termination; systematic debugging; measured iteration	2-AP-12 (combine control structures incl. nested loops); 2-AP-17 (systematically test and refine); 2-AP-19 (document programs)
3 • Conditionals (11-15)	If/then, if/then/else, compound conditions (AND/OR/NOT), sensor thresholds, multi-way decisions, tracing logic	2-AP-12 (compound conditionals and control structures); 2-AP-17 (test with a range of cases); 2-AP-13 (decompose a decision problem)
4 • Variables (16-20)	Create/name variables, update/accumulate, variables in decisions, program state, multiple variables and relationships	2-AP-11 (clearly named variables, operations on values); 2-AP-12 (variables with control structures); 2-AP-19 (document variables' roles)
5 • Functions (21-25)	Define/call functions, parameters and arguments, modular decomposition, reuse, refactoring, documentation, code review	2-AP-14 (create procedures with parameters for reuse); 2-AP-13 (decompose into subproblems); 2-AP-16 (incorporate/reuse code with attribution); 2-AP-19 (documentation); 2-AP-15 (peer feedback/review)
6 • AI & Data (26-30)	Train a classifier on collected data; model output as input; evaluate accuracy; refine with balanced data; bias, fairness, privacy, responsible use	2-DA-08 (collect data with computational tools and transform it to be reliable); 2-DA-09 (refine computational models based on their data); 2-IC-21 (bias and accessibility in technology design); 2-IC-23 (public-vs-private tradeoffs)
Pair programming (every lesson)	Driver/Navigator roles, swapping, distributed tasks, collaboration	2-AP-18 (distribute tasks and maintain a timeline when collaborating); 2-AP-15 (seek and incorporate feedback)

Cross-cutting: the debugging tasks (L7, L9, L14, L24) and test protocols reinforce **2-AP-17** and **2-AP-19**; the engineering-notebook habit and function documentation reinforce **2-AP-19**; the design-challenge reviews and the L25 code-review gallery reinforce **2-AP-15** and **2-AP-18**; the AI-impacts prompts (L8, L12, L20) and the L28-L29 data-literacy, bias, and privacy work reinforce **2-DA-08/09**, **2-IC-21**, and **2-IC-23**; the honest-AI framing throughout (a simple classifier trained on-device, never "their own ChatGPT") is maintained from the first AI Connection to the capstone.